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TECHNICAL ARCHITECTURE DOCUMENT

Salesforce CRM Platform

Northwind Bank · Sales Cloud · Financial Services Cloud · OmniStudio

REFERENCE	VERSION	DATE
TAD-NWB-CRM	1.0 — For review	28 October 2026

PREPARED BY	RECIPIENT	CLASSIFICATION
Acme Integrators	Northwind Bank	Confidential — fictional example

Document prepared by Acme Integrators for Northwind Bank. Restricted distribution Northwind Bank / Acme Integrators.

Document control

Item	Value
Title	Technical Architecture Document — Salesforce CRM Platform
Reference	TAD-NWB-CRM
Client	Northwind Bank
Integrator	Acme Integrators
Version	1.0 — For review
Date	28 October 2026
Classification	Confidential — fictional example
Scope	Salesforce organization (Production) of Northwind Bank, state as of 28 October 2026

Version history

Version	Date	Author	Purpose
1.0	28 October 2026	Acme Integrators	Consolidated version submitted to Northwind Bank for review and approval

1 Executive summary

The **Northwind Bank CRM platform** is the bank's system of engagement for retail and business customers. Designed and built by Acme Integrators on Salesforce, it supports two populations, each in a dedicated console application:

- the **branch network** (advisors and branch managers) in the « Branch Console », to manage customers, leads and loan applications;
- the **contact center** (agents and their manager) in the « Contact Center » console, to qualify leads and record contact attempts.

Customers interact indirectly through the **mobile banking app**, which creates leads through an inbound API.

The solution relies on four Salesforce foundations, complemented by specific developments:

- **Sales Cloud** carries the lead and opportunity lifecycle, from the first contact to the loan;
- **Financial Services Cloud** provides the financial account model that loan applications refer to;
- **OmniStudio** guides the loan intake journey;
- **Platform Events** decouple the loan decision from its follow-up actions.

2
Lightning applications

4
Custom business objects

12
Apex application classes, 5 test classes

4
Lightning Web Components

1 / 1
OmniScripts / Integration Procedures (active)

3
Active flows

5
Custom Metadata records (2 types)

6
Connected systems

1.1 Business capabilities covered

Domain	Main capabilities
Customer knowledge	Customer record with live positions from the core banking system (Customer 360), segment kept in sync every night
Leads	Leads from the mobile app, the branches and the contact center; default branch assignment; scoring at creation
Loan applications	Intake guided by an OmniScript, automatic scoring from configurable rules, decision, document archiving

Domain	Main capabilities
Customer notifications	SMS reminders for missing documents, follow-up tasks when a status changes
Contact center	Lead queue, contact attempts and their outcome
Steering	Loan reports by status and by branch, loan pipeline dashboard
Security	Single sign-on for employees, persona-based permission set groups, role hierarchy following the network

1.2 Key achievements

Configurable loan scoring

The score of each loan application is computed by `LoanScoringService` from rules stored as `ScoringRule__mdt` records. A new criterion or a new weight is a configuration change: no code, no deployment window.

Live Customer 360

Accounts, cards and loans are read in real time from the core banking system through a named credential (`NWB_CoreBanking`). No position is copied into Salesforce, so advisors always see the figures of the system of record.

Event-driven loan follow-up

Every status change of a loan application publishes a `LoanStatusChange__e` event. Follow-up tasks, notifications and archiving subscribe to it, so new reactions are added without touching the trigger.

Secure by design integrations

All calls go through named and external credentials; the inbound lead API runs in user mode; employees log in once through the corporate identity provider, with users created by a registration handler.

1.3 Architecture principles applied

1. **Standard first**: Sales Cloud objects, Financial Services Cloud accounts, OmniStudio journeys, queues and platform events are used before any specific development.
2. **The core banking system stays the master** of customers and positions: Salesforce reads them live and keeps only what the relationship needs.
3. **Configuration in Custom Metadata**: scoring rules and integration timeouts are records, changed without code.
4. **One trigger per object**, delegating to a handler class; services and selectors separate business logic from data access.
5. **Asynchronous for anything slow**: archiving, SMS and nightly synchronisation run outside the user transaction.
6. **Credentials outside the code**: every endpoint is a named credential; authentication is held by external credentials.

2 Purpose, scope and references

2.1 Purpose of the document

This document describes the technical architecture of the Northwind Bank CRM platform as implemented. It is the reference for:

- the understanding of the solution by the bank's teams;
- run, corrective and evolutive maintenance;
- architecture and security reviews;
- knowledge transfer between Acme Integrators, Northwind Bank and its partners.

It is written for architects, application owners, operations and security teams, and the developers who will make the platform evolve.

2.2 Scope

Included	Excluded
The Salesforce organization of Northwind Bank (production): data model, application components, automation, OmniStudio, user interface, integrations seen from Salesforce, access model, delivery and operations	Business data; internals of the core banking system, the SMS gateway, the document store, the identity provider and the mobile app; internals of the Financial Services Cloud and OmniStudio packages

2.3 References

Ref.	Document	Purpose
DR-01	Metadata of the Northwind Bank organization, retrieved in read-only mode	Source of every inventory and figure of this document
DR-02	Salesforce Well-Architected (architect.salesforce.com)	Reading grid of chapter 13
DR-03	Salesforce Integration Patterns and Practices	Classification of the integration flows

> **NOTE**
>
> Northwind Bank and Acme Integrators are fictional. This example shows what TADForce produces on a small org.

2.4 Method

This document describes the architecture **as implemented** in the Salesforce organization of Northwind Bank (Production), as of 28 October 2026. It was produced from the metadata of the organization, retrieved in read-

only mode, and from the inputs provided by Northwind Bank and Acme Integrators.

- Inventories and counts cover the components built for Northwind Bank. Components delivered by managed packages and unused standard samples are excluded.
- Each statement relies on the metadata, on a document provided by the project, or is explicitly marked as to be confirmed.
- Authentication parameters (identifiers, secrets, certificates) and business data are never reproduced.

The following components exist in the organization but are outside the documented architecture:

Type	API name	Reason
apex	`DemoDataLoader`	Demonstration data loader
flow	`Old_Branch_Survey`	Draft survey never activated
lwc	`demoCarousel`	Demonstration component, not placed in production pages

2.5 Reading conventions

- Technical names (objects, fields, classes, flows, components) are written in monospace, for example `Account`. Business labels are written in quotes, for example « Customer ».
- Salesforce terms in common use (Lead, Queue, Record Type, Permission Set, Named Credential…) are kept as is.
- Diagrams share one colour code:

Actors: users, populations, channels

Salesforce components

Systems outside the CRM

Data: objects, metadata, reference data

3 Context and vision

3.1 Business drivers

- Give every advisor a complete and current view of the customer, without leaving the CRM.
- Turn every contact — branch, contact center, mobile app — into a qualified lead handled by the right branch.
- Shorten the time from loan request to decision, with a score computed the same way everywhere.
- Keep customers informed at each step of their loan application.
- Steer the loan pipeline by status and by branch.

3.2 Actors and user populations

Population	Application	Main activities
Branch advisors	« Branch Console »	Customer 360, leads of the branch, loan applications, contact attempts
Branch managers	« Branch Console »	Same as advisors, plus branch management and review of submitted loan applications
Contact center agents	« Contact Center »	Lead qualification from the contact center queue, contact attempts
Mobile app customers	Mobile banking app	Request a call back or a loan: a lead is created through the API

3.3 Functional scope

Domain	Covered by
Leads and qualification	§ 5.2
Loan applications	§ 5.3
Customer 360	§ 5.4
Notifications	§ 5.5
Contact center	§ 5.6

3.4 Key figures

Area	Count
Lightning apps	2
Lightning pages	3
Lightning Web Components	4
Apex classes	12

Area	Count
Apex test classes	5
Apex triggers	2
Flows (active / total)	3 / 3
OmniScripts (active)	1
Integration Procedures (active)	1
Data Mappers	1
Custom objects	4
Custom fields	30
Custom Metadata types / records	2 / 5
Permission set groups / permission sets	3 / 5
Roles	10
Named / External credentials	3 / 2
Apex REST services	1
Reports / dashboards	2 / 1

4 General architecture

4.1 Context view

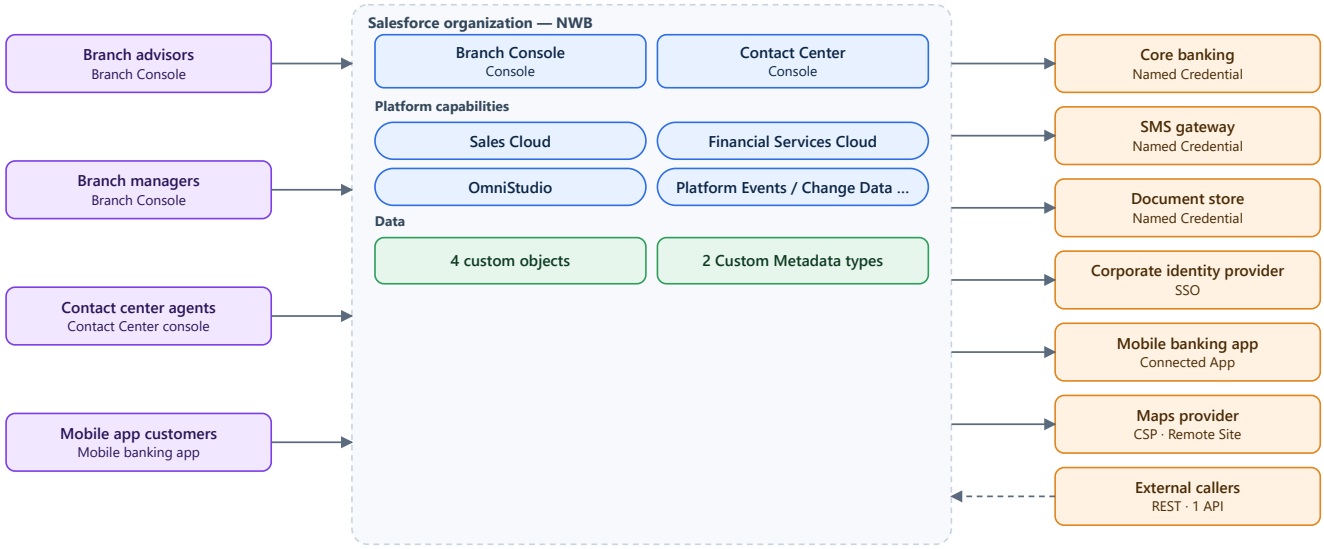


Figure 1 — Context view: users, Salesforce organization and connected systems

Users. Employees reach the platform through two Lightning console applications and log in with the corporate identity provider. Customers never log in to Salesforce: the mobile banking app calls an Apex REST service on their behalf.

Salesforce. The organization combines Sales Cloud objects, the financial account model of Financial Services Cloud, an OmniStudio journey for loan intake and platform events for the loan follow-up. Specific objects hold branches, loan applications, loan documents and contact attempts.

Connected systems. The core banking system is read in real time and feeds the nightly customer synchronisation; the SMS gateway sends reminders; the document store archives the documents of decided loan applications; the maps provider serves map tiles to the console.

4.2 Salesforce foundation and products

Product	Evidence in the metadata
Sales Cloud	object `Opportunity`, object `Lead`
Financial Services Cloud	namespace `FinServ`
OmniStudio	dedicated metadata
Platform Events / Change Data Capture	dedicated metadata

Product	Use in the solution
Sales Cloud	Leads, conversion, opportunities linked to loan applications

Product	Use in the solution
Financial Services Cloud	Financial account model referenced by loan applications (`LoanApplication__c.FinancialAccount__c`)
OmniStudio	Guided loan intake (OmniScript), server-side retrieval of positions (Integration Procedure), loan data extraction (Data Mapper)
Platform Events	Decoupling of the loan status change from its consequences

Everything else — branches, loan applications, documents, scoring, Customer 360 — is specific and documented in chapters 6 to 9.

4.3 Logical architecture in layers

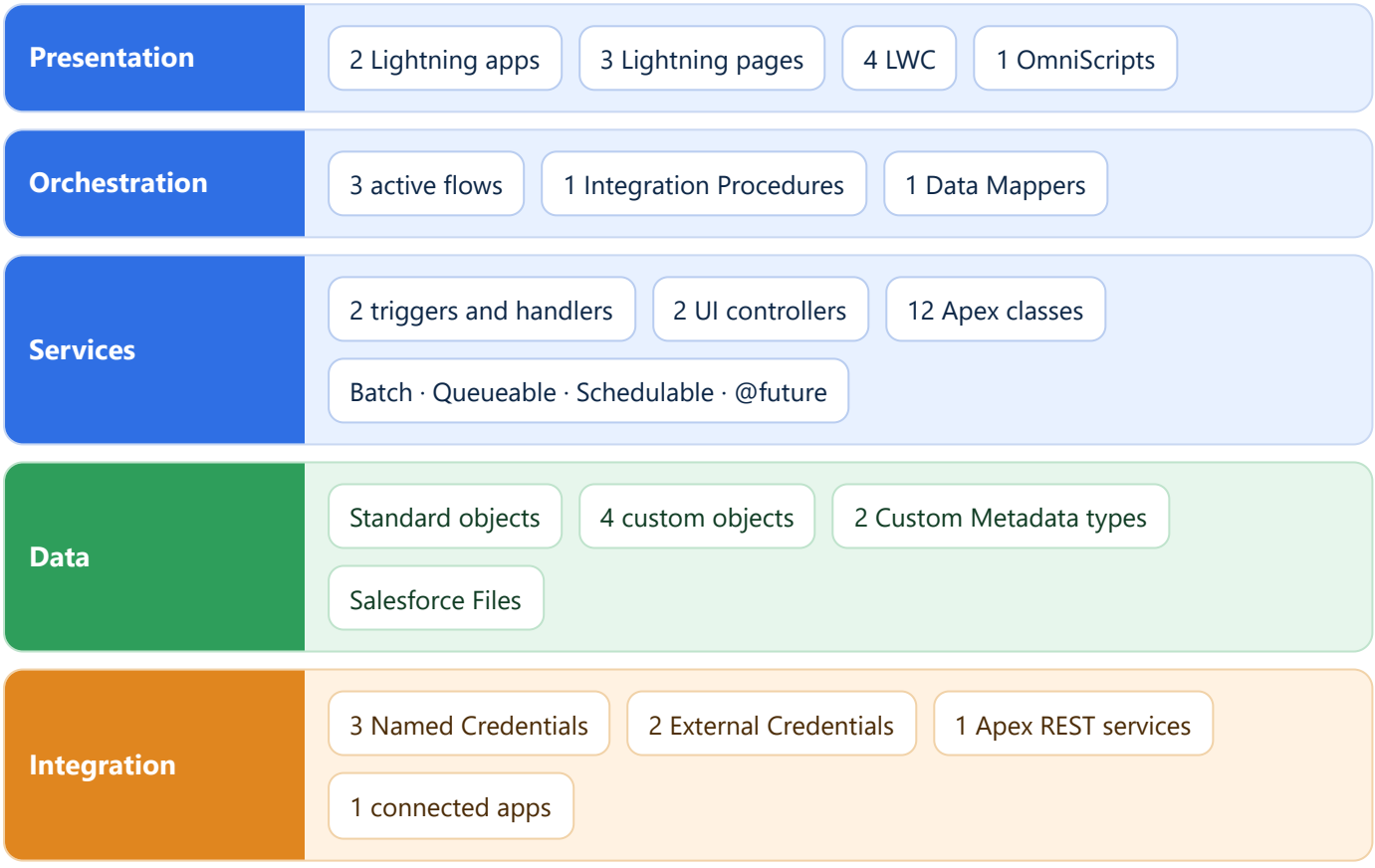


Figure 2 — Logical architecture in layers

Layer	Components	Responsibility
Presentation	Console applications, Lightning pages, LWC, OmniScript	Show information and guide the user; no heavy business rule in the browser
Orchestration	Flows, Integration Procedure, Data Mapper	Chain the steps of a process; declarative automation
Apex services	Triggers and handlers, controllers, services, selectors, asynchronous jobs	Business rules, data access, integrations, volume processing
Data	Standard and industry objects, specific objects, Custom Metadata	Business data and configuration

Layer	Components	Responsibility
Integration	Named and external credentials, Apex REST, connected app	Isolate and secure the exchanges with the bank's systems

4.4 Design principles

- **Standard first**: standard objects, queues, record types, platform events and OmniStudio before custom code.
- **Configuration over code**: business parameters in Custom Metadata (`ScoringRule__mdt`, `IntegrationSetting__mdt`).
- **Bulk-safe**: triggers process lists, the nightly synchronisation runs in batches of 50 records (`CustomerSyncBatch`).
- **Secure server-side**: controllers run `with sharing` and query in user mode; the inbound API inserts records as the calling user.
- **Loose coupling**: the trigger publishes an event; subscribers react independently.

4.5 Naming conventions

Family	Observed convention	Occurrences
Apex	*Test	5
Apex	*Service	2
Apex	*TriggerHandler	2
Apex	*Batch	1
Apex	*Client	1
Apex	*Controller	1
Apex	*Handler	1
Apex	*Helper	1
Apex	*Queueable	1
Apex	*Resource	1
Flows	Loan_*	2
PermissionSets	NWB_*	5

- Apex classes carry their role as a suffix: `*TriggerHandler`, `*Service`, `*Selector`, `*Controller`, `*Batch`, `*Queueable`, `*Client`, `*Resource`; test classes end with `Test`.
- Permission sets and permission set groups start with the bank code `NWB_`; persona groups with `NWB_Persona_`.
- Flows start with the object they work on (`Lead_`, `Loan_`).
- LWC names are in camel case and describe the component (`customer360`, `loanTimeline`).

5 Functional architecture and business processes

5.1 Functional domains

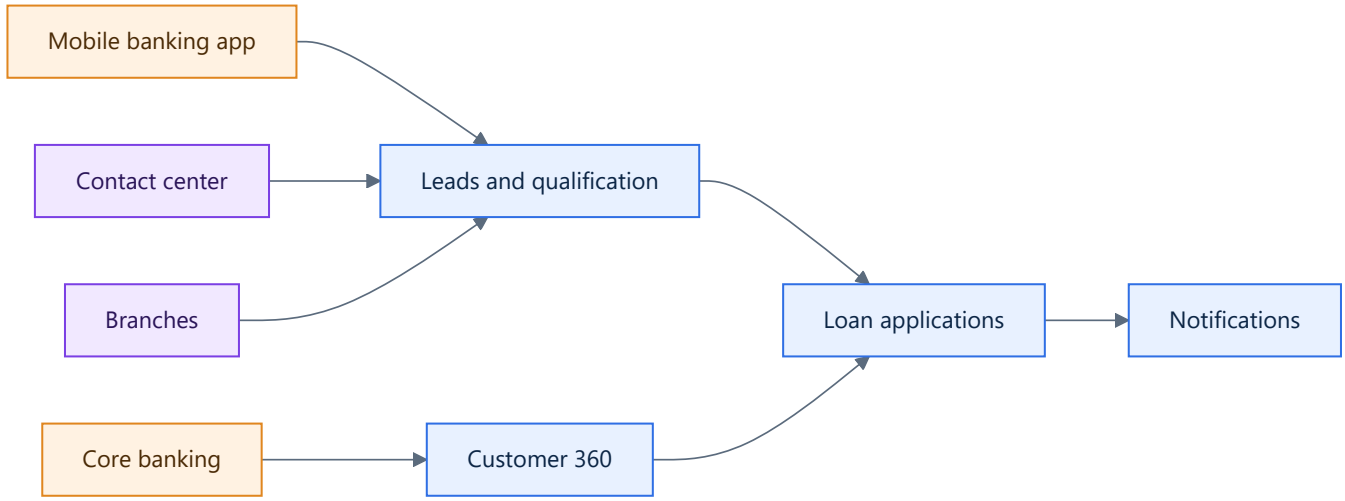


Figure 3 — Functional domains and their sequence, from the first contact to the loan

Domain	Main objects	Main components	Section
Leads and qualification	`Lead`, `ContactAttempt__c`	`LeadApiResource`, `LeadTrigger`, `Lead_Assign_Branch`	§ 5.2
Loan applications	`LoanApplication__c`, `LoanDocument__c`	`NWB_LoanIntake_English_1`, `LoanApplicationTrigger`, `LoanScoringService`	§ 5.3
Customer 360	`Account`	`customer360`, `Customer360Controller`, `CoreBankingClient`	§ 5.4
Notifications	`LoanStatusChange__e`	`Loan_Status_Notification`, `Loan_Document_Reminder`, `SmsNotificationService`	§ 5.5
Contact center	`Lead`, `ContactAttempt__c`	Queue `Contact_Center_Leads`, « Contact Center » console	§ 5.6

5.2 Leads and qualification

Leads come from three channels: the mobile banking app, the branches and the contact center.

- The mobile app calls the Apex REST service `LeadApiResource` (`/leads/v1/*`). The lead is inserted in user mode with its channel (`Channel__c`).
- At creation, `LeadTrigger` delegates to `LeadTriggerHandler`, which initialises the score (`Score__c`).
- The record-triggered flow `Lead_Assign_Branch` then assigns the lead to its branch (`Branch__c`).
- Leads waiting for the contact center are owned by the queue `Contact_Center_Leads`.
- A converted lead can no longer change status: the validation rule `Lead_Locked_When_Converted` enforces it.

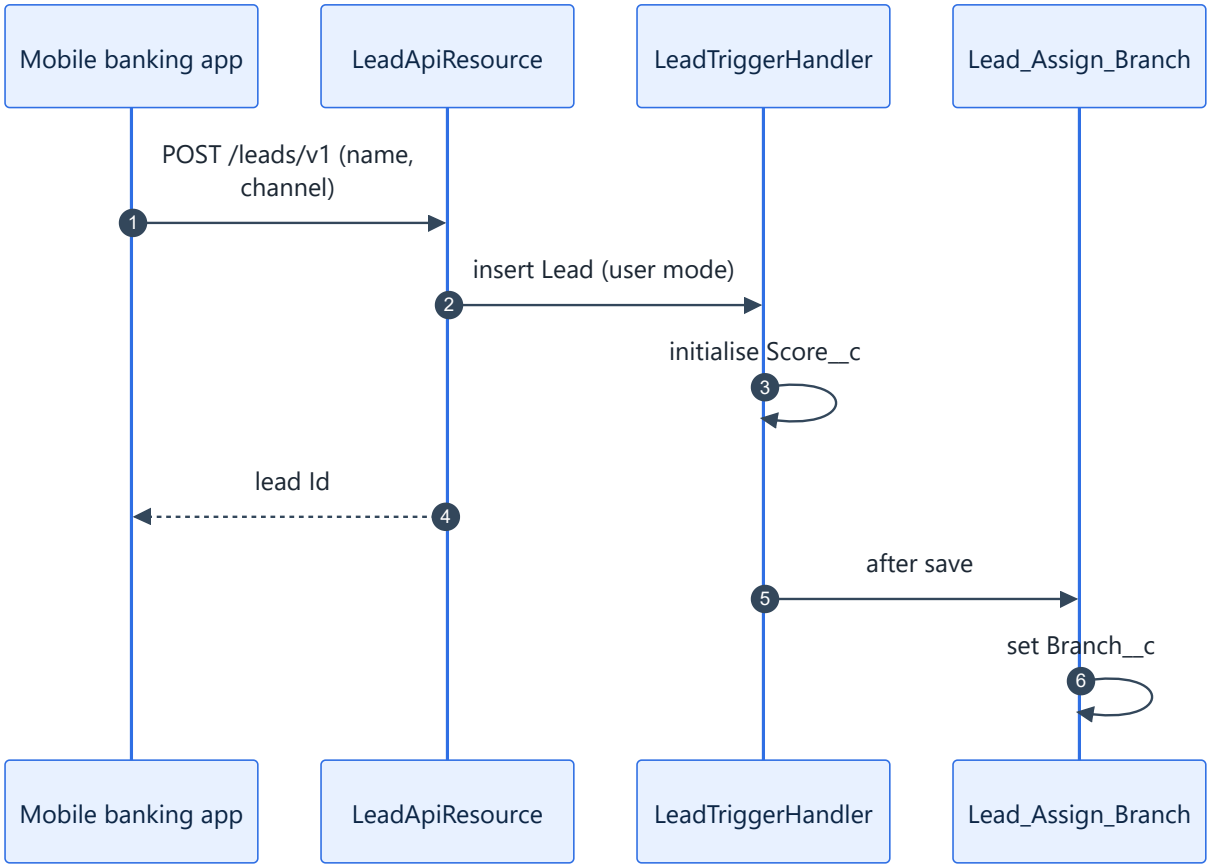


Figure 4 — Creation of a lead from the mobile banking app

5.3 Loan applications

A loan application (`LoanApplication__c`) is created from the customer record (quick action `Account.New_Loan_Application`), from the « New loan » tab (`loanWizard`) or through the guided OmniScript `NWB_LoanIntake_English_1` . Its record type distinguishes consumer loans from mortgages.

- **Scoring.** Before each save, `LoanApplicationTriggerHandler` asks `LoanScoringService` to compute `Score__c` from the `ScoringRule__mdt` rules.
- **Review.** Submitted applications are shared in read-only mode with the branch managers (sharing rule `Share_Submitted_With_Review_Team`); the queue `Loan_Review` holds the applications waiting for a reviewer.
- **Decision.** Each status change publishes a `LoanStatusChange__e` event. When the status becomes « Approved » or « Rejected », the documents are archived by `DocumentArchiveQueueable` .
- **Documents.** Each document is a `LoanDocument__c` record, child of its application (master-detail), with its archive identifier (`ArchiveId__c`).

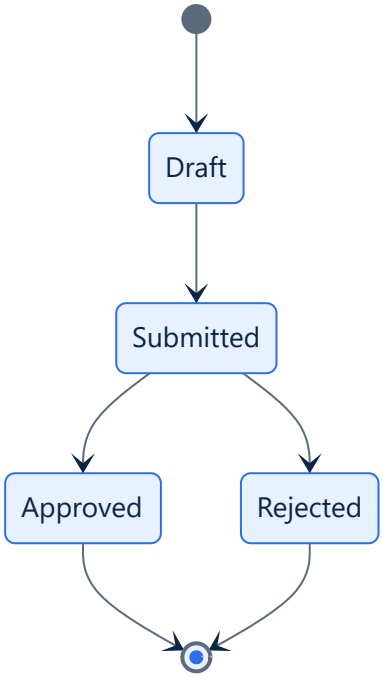


Figure 5 — Statuses of a loan application (values to be confirmed with the business team)

> **ACME INTEGRATORS CONTRIBUTION**
>
> Scoring rules are data, not code: the business team adjusts weights and criteria in `ScoringRule__mdt` without a release.

5.4 Customer 360

On the customer record page (`Account_Record_Page`), the `customer360` component displays the customer's positions read live from the core banking system:

1. the component calls `Customer360Controller.getPositions` through a cached wire adapter;
2. the controller reads the customer number (`NWB_CustomerNumber__c`) in user mode;
3. `CoreBankingClient` calls the core banking API through the named credential `NWB_CoreBanking`;
4. each position is rendered by `positionTile`.

Each night, `CustomerSyncBatch` refreshes the customer segment (`Segment__c`) from the same API.

5.5 Notifications

- **Status change.** The flow `Loan_Status_Notification` subscribes to `LoanStatusChange__e` and creates a follow-up task for the owner.
- **Missing documents.** The scheduled flow `Loan_Document_Reminder` runs daily on loan applications and calls the invocable action of `SmsNotificationService`, which sends the SMS through the SMS gateway asynchronously.

5.6 Contact center

Agents work in the « Contact Center » console on the leads of the queue `Contact_Center_Leads`. Each call is recorded as a contact attempt (`ContactAttempt__c`) linked to the lead or the customer, with its outcome (`Outcome__c`).

6 Data architecture

6.1 Modeling principles

- Standard objects carry the relationship: `Account` for customers (retail and business record types), `Lead`, `Opportunity`.
- Financial accounts follow the Financial Services Cloud model; loan applications refer to them.
- Specific objects exist only for what the standard does not cover: branches, loan applications and their documents, contact attempts.
- Every record fed by or sent to another system carries an external ID used as matching key.
- Relationships are lookups, except the documents of a loan application (master-detail: they live and die with it).
- Business parameters are Custom Metadata records.

6.2 Overview

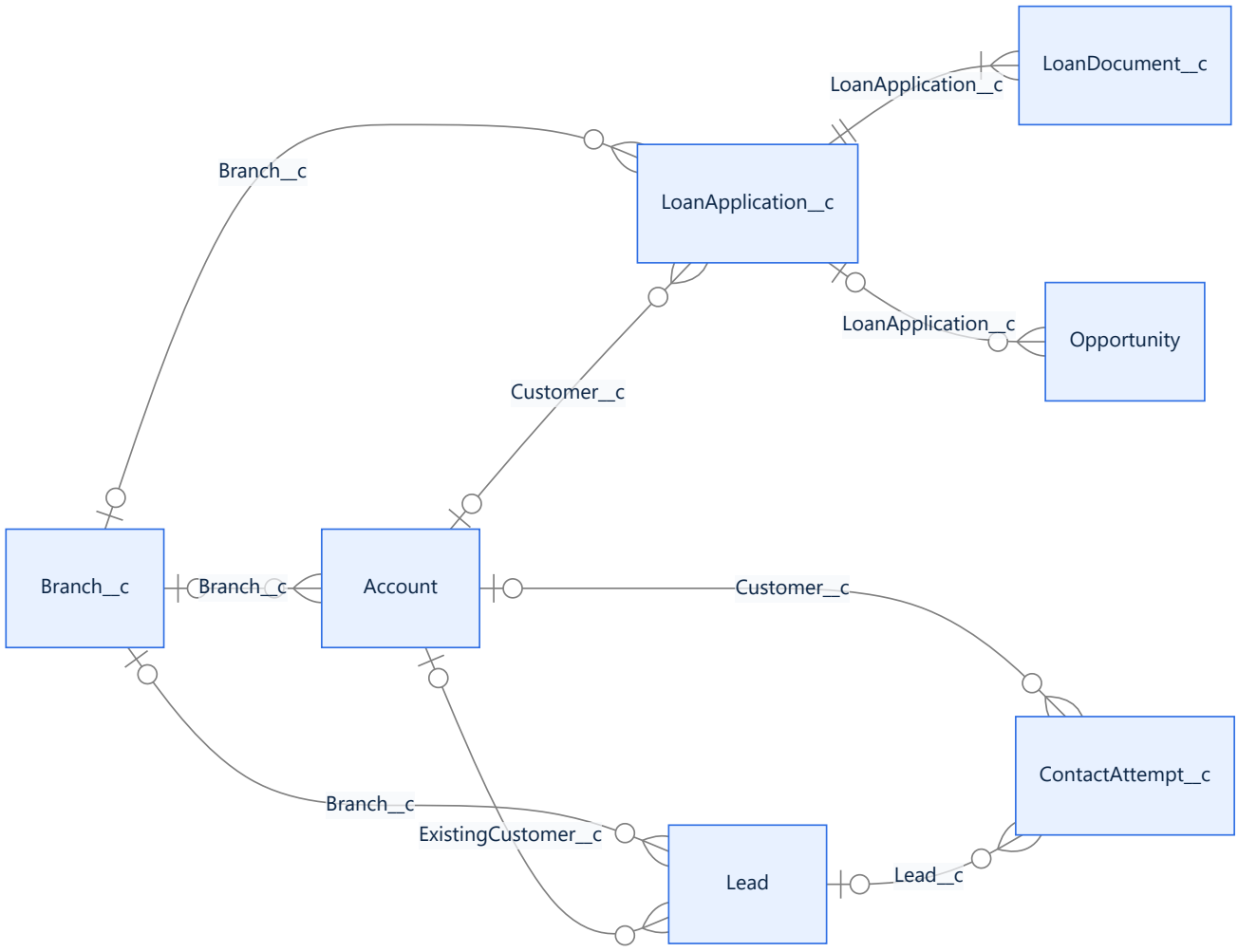


Figure 6 — Main objects and their relationships

The customer (`Account`) is the centre of the model: its branch, its loan applications, its leads and its contact attempts hang off it. The branch (`Branch__c`) owns customers and leads by default. A loan application gathers its documents and may be linked to an opportunity and to a financial account.

6.3 Objects by domain

6.3.1 Customers and branches

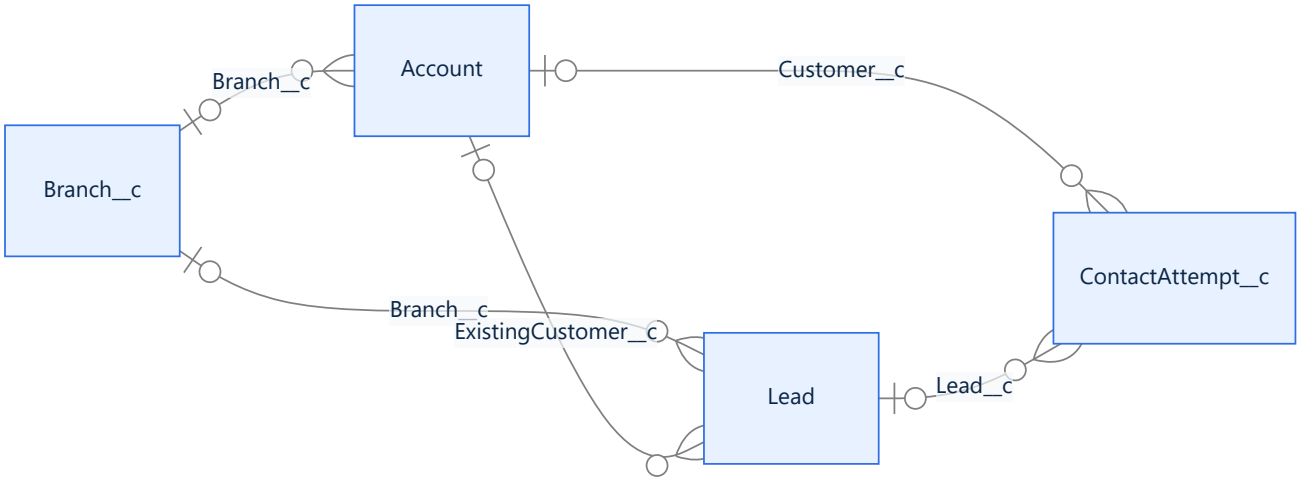


Figure 7 — Customers, branches and leads

Object	Role	Key fields
`Account`	Customer (retail or business)	`NWB_CustomerNumber__c` (core banking key), `Segment__c` , `Branch__c` , `Advisor__c`
`Branch__c`	Branch of the network	`BranchCode__c` , `Region__c` , `Manager__c`
`Lead`	Prospect or customer request	`Channel__c` , `Score__c` , `ExistingCustomer__c` , `Branch__c`
`ContactAttempt__c`	Call or contact by an advisor or an agent	`Customer__c` , `Lead__c` , `Outcome__c`

6.3.2 Loans

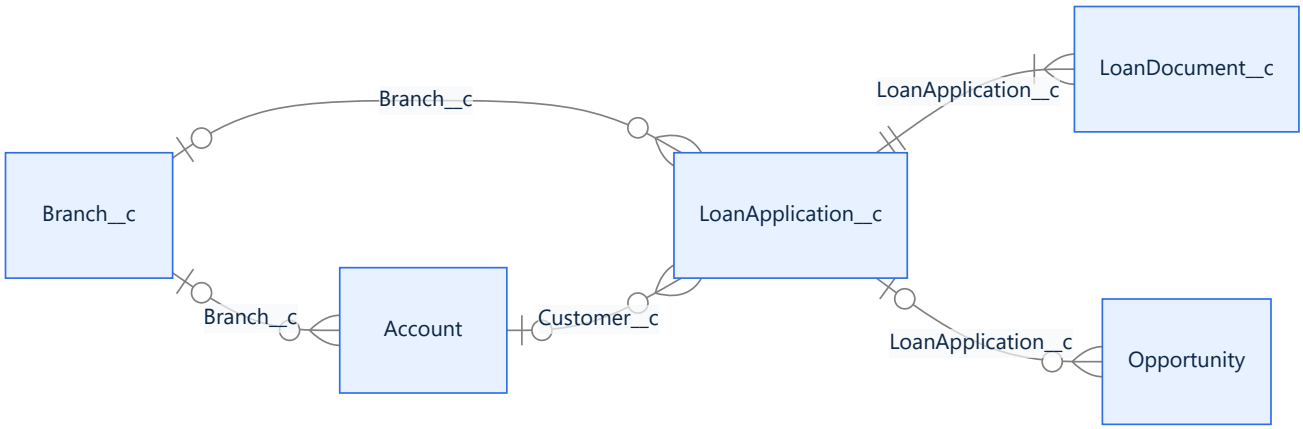


Figure 8 — Loan applications and their documents

Object	Role	Key fields
`LoanApplication__c`	Loan request from intake to decision	`Customer__c`, `Status__c`, `Amount__c`, `Score__c`, `CoreBankingId__c`, `FinancialAccount__c`
`LoanDocument__c`	Document of an application	`LoanApplication__c` (master), `DocumentType__c`, `ArchiveId__c`
`Opportunity`	Commercial opportunity of the loan	`LoanApplication__c`

6.4 Industry cloud objects

Object	Use
`FinServ__FinancialAccount__c`	Financial account (loan, current account) to which a loan application can be attached through `FinancialAccount__c`

The rest of the Financial Services Cloud model is not referenced by the specific metadata.

6.5 External IDs and integration keys

Object	Field
`Account`	`NWB_CustomerNumber__c`
`Branch__c`	`BranchCode__c`
`LoanApplication__c`	`CoreBankingId__c`
`LoanDocument__c`	`ArchiveId__c`

- `NWB_CustomerNumber__c` and `CoreBankingId__c` are the keys of the core banking system, used by the nightly synchronisation and by the Customer 360 calls.
- `BranchCode__c` identifies a branch in the bank's referential.

- `ArchiveId__c` is the identifier returned by the document store.

6.6 Record types and business processes

Object	Record types
`Account`	`Business_Customer`, `Retail_Customer`
`LoanApplication__c`	`Consumer_Loan`, `Mortgage`

Customer record types separate retail and business customers (different page layouts and segments). Loan record types separate consumer loans and mortgages, which follow different scoring rules and documents.

6.7 Configuration in Custom Metadata

Type	Records
`IntegrationSetting__mdt`	2
`ScoringRule__mdt`	3

Type	What it configures
`ScoringRule__mdt`	Criteria and weights of the loan score (`Criterion__c`, `Weight__c`)
`IntegrationSetting__mdt`	Timeouts of the outbound integrations (`TimeoutMs__c`)

6.8 Platform events and change data capture

API name	Label
`LoanStatusChange__e`	Loan status change

`LoanStatusChange__e` carries the loan application identifier and its new status. It is published by `LoanApplicationTriggerHandler` and consumed by the flow `Loan_Status_Notification`. No change data capture entity is selected.

6.9 Files and documents

The documents of a loan application are described by `LoanDocument__c` records; their content is archived in the document store once the application is decided, and the archive identifier is kept on the record.

6.10 Data lifecycle

Field history tracking is enabled on the specific objects. No archiving or purge policy is configured in Salesforce for loan applications and contact attempts: the retention rules are to be confirmed with the bank.

7 Application architecture

7.1 Apex code architecture

Nature	Count
Batch	1
Queueable	1
Schedulable	1
@future	1
REST	1
Invocable	1
LWC/Aura controller	2
Callout	3
SSO registration handler	1
Global	2

7.1.1 Design principles

- One trigger per object, with no logic in the trigger body: each trigger calls the `run()` method of its handler.
- Business rules live in services (`LoanScoringService`), data access in selectors (`LoanApplicationSelector`), external calls in clients (`CoreBankingClient`).
- Code processes lists of records, never one record at a time.
- 8 classes declare `with sharing` ; queries from user-facing code run `WITH USER_MODE` .

7.1.2 Layers and conventions

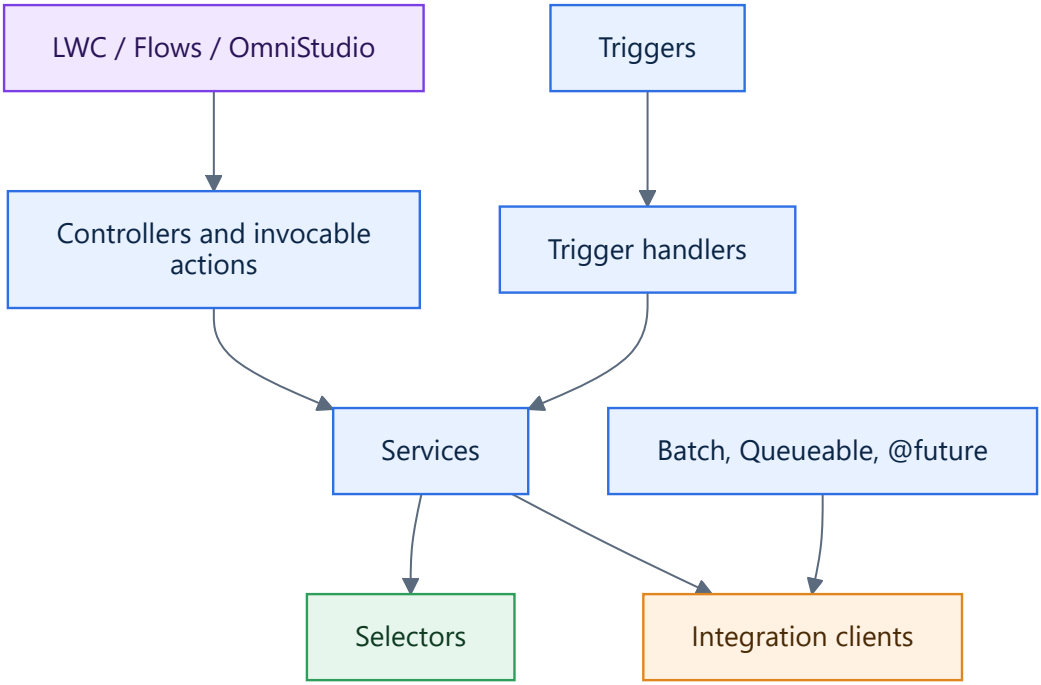


Figure 9 — Layers of the Apex code

Layer	Convention	Examples
Trigger handlers	`<Object>TriggerHandler`	`LoanApplicationTriggerHandler`, `LeadTriggerHandler`
Services	`<Domain>Service`	`LoanScoringService`, `SmsNotificationService`
Selectors	`<Object>Selector`	`LoanApplicationSelector`
Controllers	`<Component>Controller`	`Customer360Controller`
Integration clients	`<System>Client`	`CoreBankingClient`
Asynchronous jobs	`*Batch`, `*Queueable`	`CustomerSyncBatch`, `DocumentArchiveQueueable`
Exposed services	`*Resource`	`LeadApiResource`

7.1.3 Classes by functional domain

Domain	Main classes	Role
Loans	`LoanApplicationTriggerHandler`, `LoanScoringService`, `LoanApplicationSelector`, `DocumentArchiveQueueable`	Scoring, events, archiving
Customer 360	`Customer360Controller`, `CoreBankingClient`, `CustomerSyncBatch`	Live positions, nightly synchronisation
Leads	`LeadApiResource`, `LeadTriggerHandler`	Inbound API, initialisation
Notifications	`SmsNotificationService`	SMS reminders
Access	`SsoRegistrationHandler`	User creation at single sign-on

7.2 Triggers

Object	Trigger	Handler	Events
`LoanApplication__c`	`LoanApplicationTrigger`	`LoanApplicationTriggerHandler`	before insert, before update, after update
`Lead`	`LeadTrigger`	`LeadTriggerHandler`	before insert

7.2.1 Order of execution

On `Lead`, the trigger runs before save (score initialisation); the flow `Lead_Assign_Branch` runs after save (branch assignment). On `LoanApplication__c`, scoring happens before save, then the event is published and the archiving job is enqueued after update.

7.3 Cross-cutting mechanisms

Loan status propagation. A single place detects status changes and publishes them; every consequence is a subscriber.

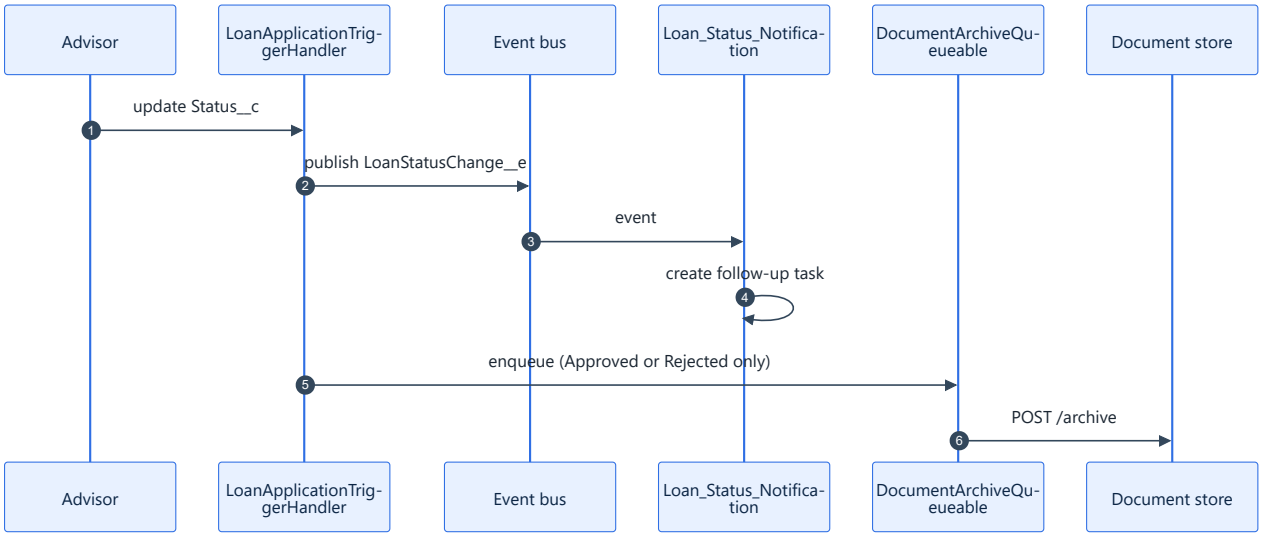


Figure 10 — Propagation of a loan status change

7.4 Asynchronous and scheduled processing

Job	Type	Trigger or schedule	Purpose
`CustomerSyncBatch`	Batch and Schedulable (batches of 50)	Scheduled (schedule to be confirmed)	Refresh the customer segment from the core banking system
`DocumentArchiveQueueable`	Queueable with callouts	Status « Approved » or « Rejected »	Archive the documents of the application
`SmsNotificationService.send`	`@future` with callout	Invocable action of `Loan_Document_Reminder`	Send the SMS reminders

7.5 Flows

Flows handle simple record automation and scheduling; Apex handles scoring, integrations and volume.

Flow	Type	Object	Purpose
`Lead_Assign_Branch`	Record-triggered, after save, on create	`Lead`	Assign the branch of the lead
`Loan_Document_Reminder`	Scheduled, daily	`LoanApplication__c`	Remind customers of missing documents by SMS
`Loan_Status_Notification`	Platform event-triggered	`LoanStatusChange__e`	Create a follow-up task

The full list is in Appendix B — Triggers and flows.

7.6 OmniStudio

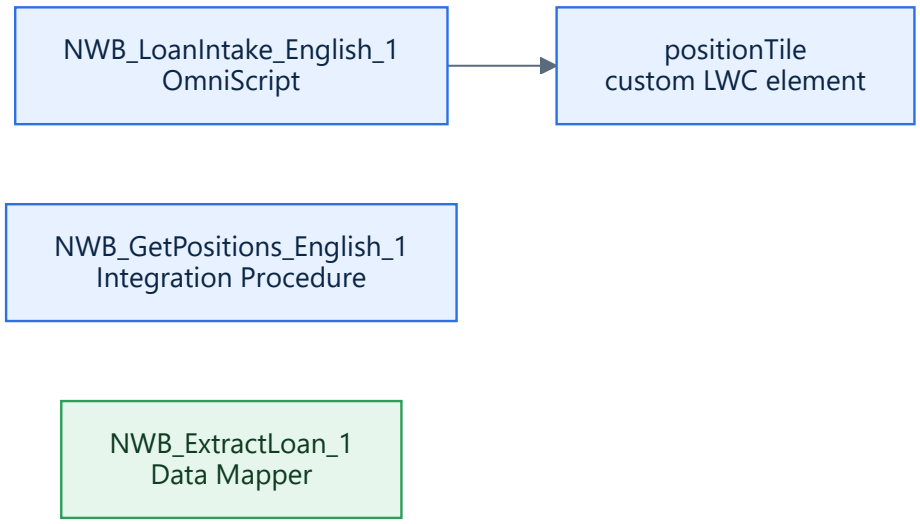


Figure 11 — OmniStudio components of the loan intake

Component	Type	Role
`NWB_LoanIntake_English_1`	OmniScript (`NWB/LoanIntake`)	Guided loan intake; reuses the `positionTile` component to show the customer's positions
`NWB_GetPositions_English_1`	Integration Procedure	Server-side retrieval of positions (steps to be confirmed)
`NWB_ExtractLoan_1`	Data Mapper (extract)	Extraction of loan data (fields to be confirmed)

7.7 Key sequences

Nightly customer synchronisation.

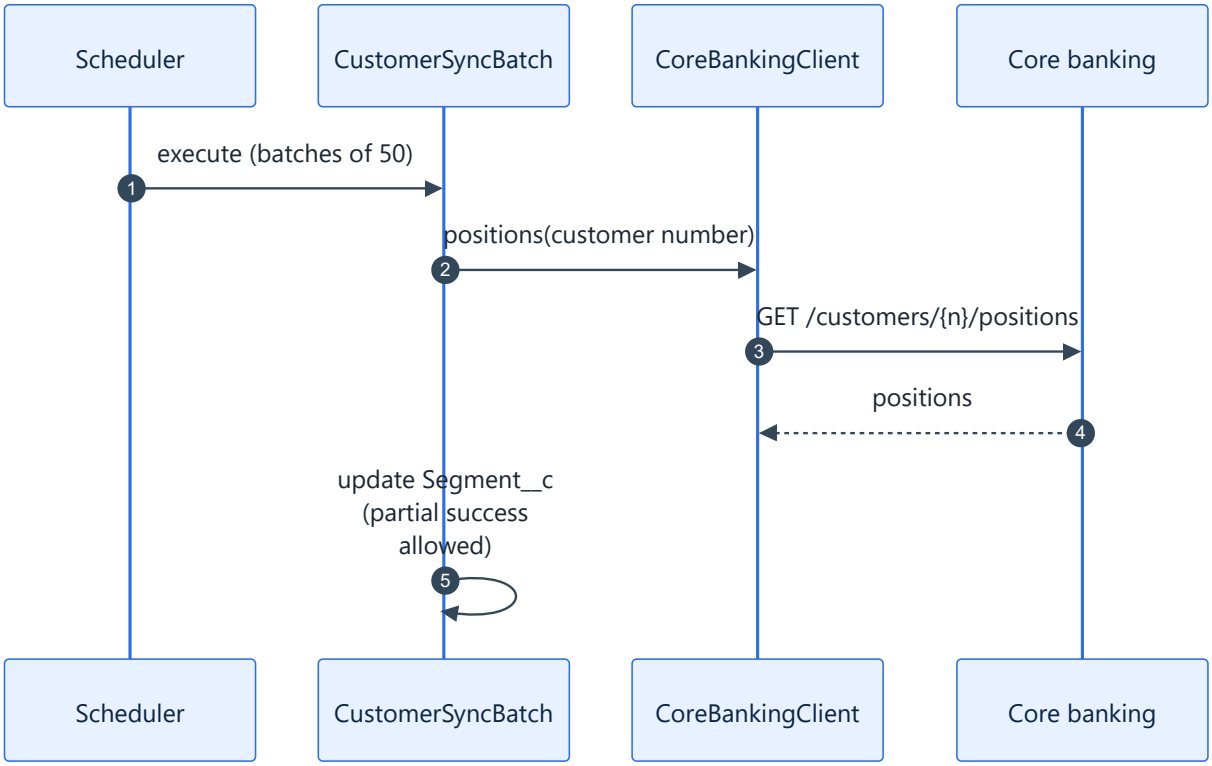


Figure 12 — Nightly synchronisation of the customer segment

8 User interface

8.1 Lightning apps and navigation

Both applications are Lightning consoles for desktop:

- « Branch Console » (`NWB_Branch_Console`): customers, leads, loan applications, branches and the « New loan » tab; its home page is `Branch_Home` ;
- « Contact Center » (`NWB_Contact_Center`): leads, customers and contact attempts.

8.2 Lightning pages

Page	Object	Custom components
`Account_Record_Page`	`Account`	`customer360`
`Loan_Application_Record_Page`	`LoanApplication__c`	`loanTimeline`
`Branch_Home`	Home page	Standard components only

8.3 Lightning Web Components

8.3.1 By domain

Domain	Components	Role
Customer 360	`customer360` , `positionTile`	Live positions of the customer
Loans	`loanTimeline` , `loanWizard`	Timeline of an application; creation wizard in a tab

8.4 Communication between components

- `customer360` reads its data with a cached wire adapter on `Customer360Controller` and passes each position to `positionTile` as a public property.
- `loanTimeline` uses Lightning Message Service; the channel `LoanSelected` is defined for selection events between components.
- `loanWizard` uses the navigation service to open the created application.

9 Integration architecture

9.1 Integration principles

- The core banking system remains the system of record: Salesforce reads positions live and stores only the keys and the segment.
- Every outbound call uses a named credential; authentication is held by an external credential, never by code.
- Calls from triggers and user transactions are asynchronous (queueable, `@future`); interactive reads are synchronous with an explicit timeout.
- The inbound API creates records in user mode, with the permissions of the integration user.

9.2 Integration patterns

Pattern	Flows
Remote Process Invocation — Request and Reply	FLX-01 Customer positions
Remote Process Invocation — Fire and Forget	FLX-03 Document archiving, FLX-04 SMS reminders
Batch Data Synchronization	FLX-02 Nightly customer synchronisation
Remote Call-In	FLX-05 Lead creation from the mobile app

9.3 Flow map

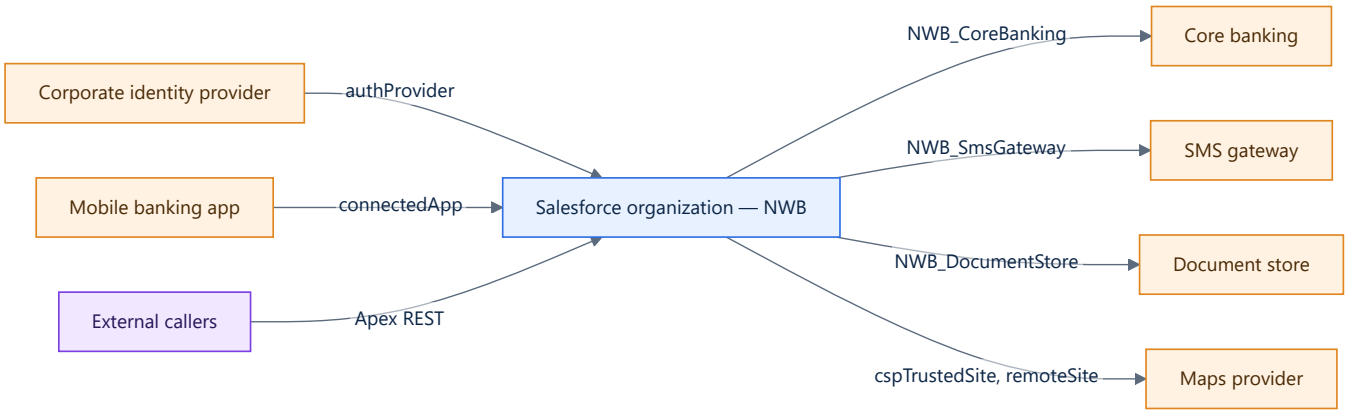


Figure 13 — Integration map: connected systems and Salesforce endpoints

ID	Flow	Direction	System	Salesforce component	Pattern
FLX-01	Customer positions	Outbound	Core banking	`CoreBankingClient`	Request and Reply
FLX-02	Customer segment synchronisation	Outbound	Core banking	`CustomerSyncBatch`	Batch Data Synchronization
FLX-03	Document archiving	Outbound	Document store	`DocumentArchiveQueueable`	Fire and Forget

ID	Flow	Direction	System	Salesforce component	Pattern
FLX-04	SMS reminders	Outbound	SMS gateway	`SmsNotificationService`	Fire and Forget
FLX-05	Lead creation	Inbound	Mobile banking app	`LeadApiResource`	Remote Call-In
FLX-06	Employee login	Inbound	Corporate identity provider	`NWB_Corporate_IdP`, `SsoRegistrationHandler`	Single sign-on

9.4 Connected systems

System	Description	Salesforce endpoints
Core banking	System of record for customers, accounts, contracts and positions	`NWB_CoreBanking`
SMS gateway	Telecom partner sending SMS notifications	`NWB_SmsGateway`
Document store	Long-term archive of the loan documents	`NWB_DocumentStore`
Corporate identity provider	Single sign-on for employees (OpenID Connect)	`NWB_Corporate_IdP`
Mobile banking app	Customer app that creates leads through the Salesforce API	`NWB_Mobile_Banking`
Maps provider	Map tiles and geocoding shown in the branch console	`NWB_Map_Tiles`, `NWB_Maps`

9.4.1 Core banking

Read through `NWB_CoreBanking`, a secured endpoint whose authentication is held by the external credential `NWB_CoreBanking_OAuth` (OAuth, named principal). `CoreBankingClient` calls `GET /customers/{number}/positions` with a 10-second timeout, for the Customer 360 (FLX-01) and the nightly synchronisation (FLX-02).

9.4.2 Document store

`DocumentArchiveQueueable` posts the identifiers of decided applications to `/archive` through `NWB_DocumentStore`.

9.4.3 SMS gateway

`SmsNotificationService` posts messages through `NWB_SmsGateway`, authenticated by the external credential `NWB_SmsGateway_Key`.

9.4.4 Maps provider

Map tiles are allowed in the console by the CSP trusted site `NWB_Map_Tiles`; geocoding calls are allowed by the remote site `NWB_Maps`.

9.5 Inbound APIs

URL mapping	Apex class	Nature
`/leads/v1/*`	`LeadApiResource`	Apex REST

The mobile banking app authenticates with the connected app `NWB_Mobile_Banking` (scopes `Api` and `RefreshToken`) and calls `POST /services/apexrest/leads/v1/`. The service creates a lead with the name, the channel and, when known, the customer number; the record is inserted with the permissions of the calling user.

9.6 Authentication and single sign-on

Employees log in through the authentication provider `NWB_Corporate_IdP` (OpenID Connect). At first login, `SsoRegistrationHandler` creates the user from the identity provider claims and sets its federation identifier.

9.7 Allowed domains

Kind	API name	Host
Remote Site	`NWB_Maps`	maps.geo-provider.example
CSP Trusted Site	`NWB_Map_Tiles`	tiles.geo-provider.example

Both entries serve the maps provider: tiles in the Lightning pages (CSP) and server-side calls (remote site).

10 Security and access

10.1 Principles

- Least privilege: a minimal profile, rights added by persona through permission set groups.
- Record access follows the network: private defaults, opened by the role hierarchy and by targeted sharing rules.
- Security is enforced on the server: `with sharing`, user-mode queries, inbound API in user mode.
- No secret in code or configuration files: authentication is held by external credentials.

10.2 Authentication and user provisioning

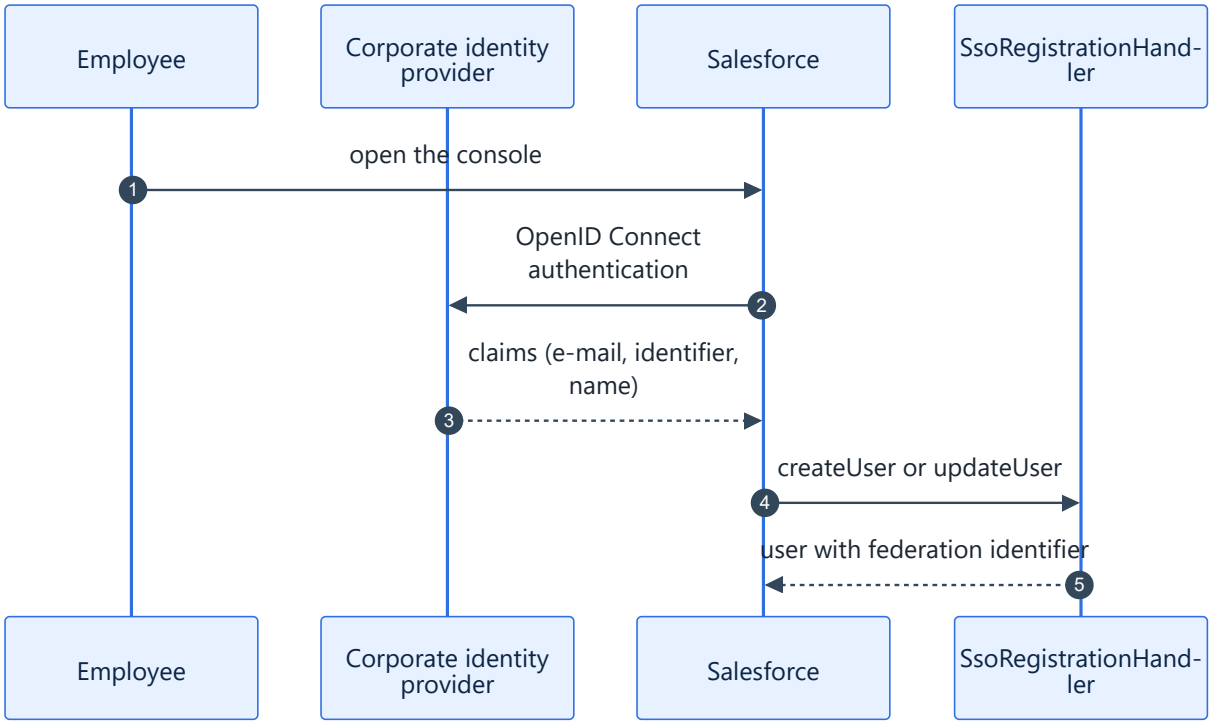


Figure 14 — Single sign-on and user creation

The assignment of permission set groups after creation is to be confirmed with the bank (manual or automated).

10.3 Personas and permission set groups

Persona	Permission set group	Main rights
Branch advisor	`NWB_Persona_Advisor`	`NWB_Advisor`: loan applications and contact attempts (create, edit), branches (read)
Branch manager	`NWB_Persona_Branch_Manager`	`NWB_Advisor` and `NWB_Branch_Manager`: branches (edit)

Persona	Permission set group	Main rights
Contact center agent	`NWB_Persona_Agent`	`NWB_Contact_Center_Agent` : contact attempts

Two technical permission sets complete the model: `NWB_Integration_API` (API access for the integration user) and `NWB_Support_Admin` (support administration).

10.4 Role hierarchy

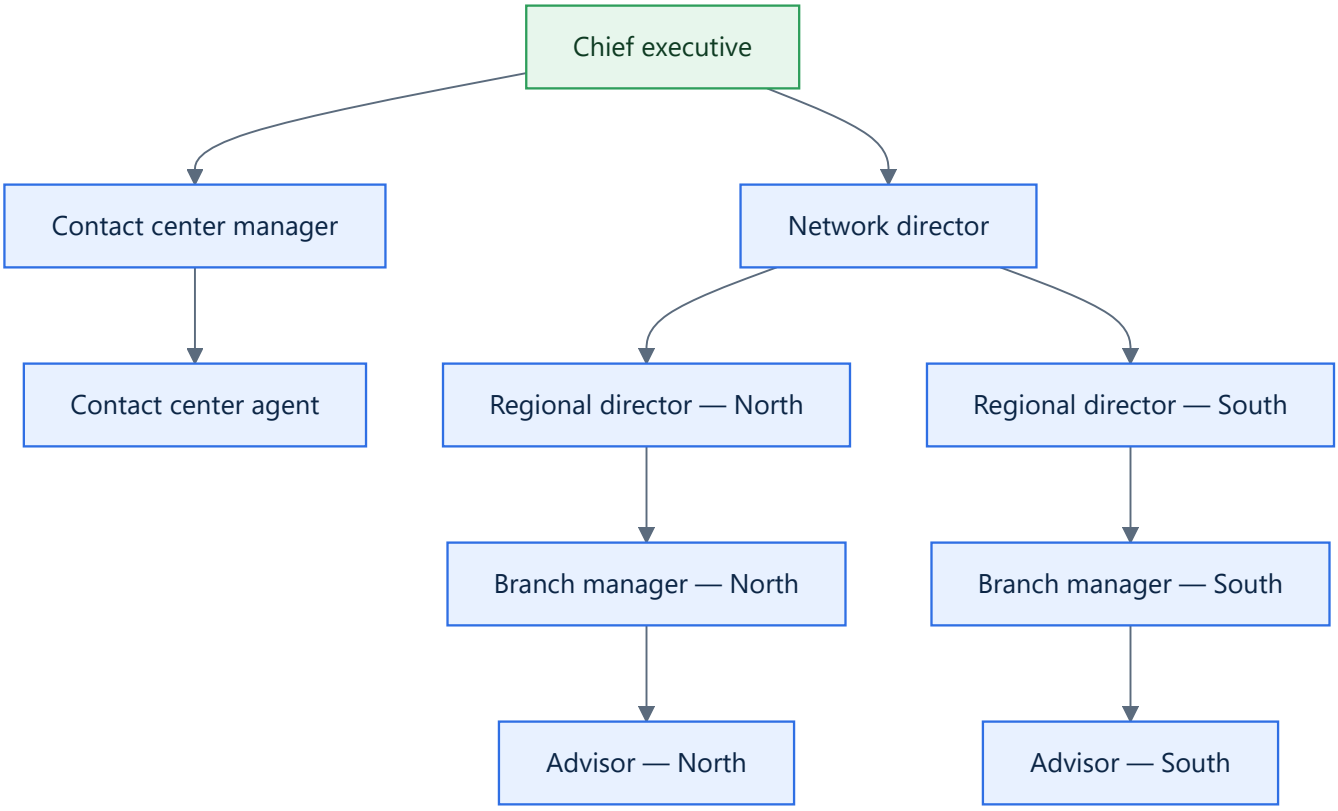


Figure 15 — Role hierarchy

The hierarchy mirrors the network: chief executive, network director, regional directors, branch managers and advisors per region, and a separate branch for the contact center.

10.5 Sharing model

10.5.1 Organization-wide defaults

Object	Default sharing (internal / external)
`Account`	Private / Private
`Branch__c`	Read / Private
`ContactAttempt__c`	Private / Private

Object	Default sharing (internal / external)
`Lead`	Private / Private
`LoanApplication__c`	Private / Private
`LoanDocument__c`	ControlledByParent / ControlledByParent
`Opportunity`	Private / Private

Customers, leads, opportunities and loan applications are private; branches are readable by all; loan documents follow their application.

10.5.2 Sharing rules and queues

Object	Rule	Kind	Access
`LoanApplication__c`	`Share_Submitted_With_Review_Team`	criteria	Read

API name	Label	Object
`Contact_Center_Leads`	Contact center leads	`Lead`
`Loan_Review`	Loan review	`LoanApplication__c`

Submitted loan applications are opened in read-only mode to the group `All_Branch_Managers`, which includes the managers' superiors. Queues hold the work not yet assigned: loan applications to review and leads for the contact center.

10.6 Application security

- 8 classes declare `with sharing` and 1 `inherited sharing`; the nightly synchronisation runs `without sharing` as a system job.
- Queries from the Customer 360 and the selectors run `WITH USER_MODE`; the inbound API inserts as user.
- Validation rules protect data consistency (`Lead_Locked_When_Converted` , `Amount_Must_Be_Positive`).

10.7 Session settings and protections

Setting	Value
Session timeout	Two hours, with forced logout
Session locked to the domain	Yes
HttpOnly cookies	Yes
Password	12 characters minimum, upper and lower case, digits and special characters; expires after 90 days; locked after 5 attempts

10.8 Integration users

The mobile banking app calls Salesforce as an integration user with the profile « NWB Integration » (Salesforce Integration licence) and the permission set `NWB_Integration_API`.

11 Reporting and analytics

11.1 Reports and dashboards

Folder	Count
`Loan_Reports`	2

The folder `Loan_Reports` holds the loan reports by status and by branch; the dashboard `Loan_Pipeline` presents the pipeline to branch managers. The branch home page embeds the dashboards.

12 Environments, delivery and operations

12.1 Environments

The environments (sandboxes and their purpose) are to be confirmed with the bank. This document describes the production organization.

12.2 Release management

The project is a Salesforce DX project (source format, API version 62.0). Metadata is deployed first, OmniStudio components second, then the active versions are checked. The pipeline and its approval gates are to be confirmed.

12.3 Scheduled processing

Processing	Frequency	What to monitor
`CustomerSyncBatch`	Nightly (to be confirmed)	Failed batches and records not updated (partial success)
`Loan_Document_Reminder`	Daily	Flow errors, SMS gateway errors

12.4 Logging and monitoring

Errors of asynchronous jobs are visible in the Apex jobs monitor and flow errors in the flow error e-mails. No specific logging framework is present in the metadata.

12.5 Test strategy

5 test classes cover scoring, the trigger handler, the nightly batch and the inbound API. Callouts are isolated behind `CoreBankingClient`, which makes them mockable. Code coverage is to be read in the organization.

13 Non-functional requirements

13.1 Measures per requirement

Requirement	Measures in place
Security	Single sign-on, persona permission set groups, private defaults, user-mode queries, external credentials
Performance	Positions read on demand with a cached wire adapter; callouts with explicit timeouts
Volume	Nightly synchronisation in batches of 50 records; asynchronous archiving and SMS
Availability	Partial success on synchronisation; Customer 360 independent from the nightly job
Maintainability	Layered Apex, naming conventions, configuration in Custom Metadata
Traceability	Field history on specific objects, status change events

13.2 Salesforce Well-Architected reading

Pillar	Strengths observed
Trusted	Credentials outside the code, server-side security, least-privilege personas
Easy	Two task-oriented consoles, guided intake, live Customer 360
Adaptable	Event-driven follow-up, scoring by configuration, clear separation of layers

14 Architecture decisions

ID	Decision	Rationale	Consequence
ADR-01	Read customer positions live from the core banking system	Single source of truth, no duplication of financial data	Customer 360 depends on the availability of the API
ADR-02	Store the scoring rules in Custom Metadata	Business changes without releases	Rules are deployable and auditable like metadata
ADR-03	Publish an event on each loan status change	Add reactions without changing the trigger	Subscribers run asynchronously
ADR-04	Expose an Apex REST service for the mobile app	Control of validation and defaults on the Salesforce side	The service is versioned in its URL (`/leads/v1/*`)

15 Outlook

- Extend the event-driven model to the contact center (notifications on lead assignment).
- Enrich the loan intake journey with document upload.
- Open a customer portal for loan follow-up.

The detailed improvement register is kept in the companion Improvement Axes document.

Appendices

Appendix A — Apex: asynchronous processing and exposed services

Apex class	Nature
`CustomerSyncBatch`	Batch, Schedulable
`DocumentArchiveQueueable`	Callout, Queueable
`SmsNotificationService`	Callout, @future

URL mapping	Apex class	Nature
`/leads/v1/*`	`LeadApiResource`	Apex REST

Appendix B — Triggers and flows

B.1 Triggers

Trigger	Object	Events
`LeadTrigger`	`Lead`	before insert
`LoanApplicationTrigger`	`LoanApplication__c`	before insert, before update, after update

B.2 Flows

API name	Label	Type	Start	Object	Status
`Lead_Assign_Branch`	Lead — assign branch	AutoLaunchedFlow	RecordAfterSave · Create	`Lead`	Active
`Loan_Document_Reminder`	Loan — document reminder	AutoLaunchedFlow	Scheduled · Daily	`LoanApplication__c`	Active
`Loan_Status_Notification`	Loan — status notification	AutoLaunchedFlow	PlatformEvent	`LoanStatusChange__e`	Active

Appendix C — Custom objects

API name	Label	Default sharing (internal / external)	Custom fields	Record types	Relationships	Description
`Branch__c`	Branch	Read / Private	3	0	1	Branch of the bank network: code, region and manager. Owner of customers and leads by default.
`ContactAttempt__c`	Contact attempt	Private / Private	3	0	2	Outbound contact attempt by an advisor or an agent, with its outcome.

API name	Label	Default sharing (internal / external)	Custom fields	Record types	Relationships	Description
`LoanApplication__c`	Loan application	Private / Private	7	2	3	Loan application from intake to decision: customer, amount, score and status.
`LoanDocument__c`	Loan document	ControlledByParent / ControlledByParent	3	0	1	—

C.1 Extended standard objects

API name	Custom fields	Record types	Default sharing (internal / external)
`Account`	4	2	Private / Private
`Lead`	4	0	Private / Private
`Opportunity`	1	0	Private / Private

C.2 Validation rules

Object	Rule	Status
`Lead`	`Lead_Locked_When_Converted`	Active
`LoanApplication__c`	`Amount_Must_Be_Positive`	Active

Appendix D — Custom Metadata types

Type	Records
`IntegrationSetting__mdt`	2
`ScoringRule__mdt`	3

Appendix E — Active OmniStudio components

E.1 OmniScripts

API name	Label	Type	Language	Status
`NWB_LoanIntake_English_1`	Loan intake	NWB/LoanIntake	English	Active

E.2 Integration Procedures

API name	Label	Type	Status
`NWB_GetPositions_English_1`	Get positions	NWB/GetPositions	Active

E.3 Data Mappers

API name	Type
`NWB_ExtractLoan_1`	Extract

Appendix F — User interface

F.1 Lightning apps

API name	Label	Navigation	Form factors
`NWB_Branch_Console`	Branch Console	Console	Large
`NWB_Contact_Center`	Contact Center	Console	Large

F.2 Lightning pages

API name	Type	Object	Custom components
`Account_Record_Page`	RecordPage	`Account`	`customer360`
`Branch_Home`	HomePage	—	—
`Loan_Application_Record_Page`	RecordPage	`LoanApplication__c`	`loanTimeline`

F.3 Lightning Web Components

API name	Placed in	Apex controllers
`customer360`	`Account_Record_Page`	`Customer360Controller`
`loanTimeline`	`Loan_Application_Record_Page`	—
`loanWizard`	`Loan_Wizard`	—
`positionTile`	`customer360`, `NWB_LoanIntake_English_1`	—

Appendix G — Access

G.1 Permission set groups

API name	Label	Permission sets
`NWB_Persona_Advisor`	Persona — advisor	`NWB_Advisor`
`NWB_Persona_Agent`	Persona — contact center agent	`NWB_Contact_Center_Agent`
`NWB_Persona_Branch_Manager`	Persona — branch manager	`NWB_Advisor`, `NWB_Branch_Manager`

G.2 Permission sets

API name	Label	License
`NWB_Advisor`	Advisor	Salesforce
`NWB_Branch_Manager`	Branch manager	Salesforce
`NWB_Contact_Center_Agent`	Contact center agent	Salesforce
`NWB_Integration_API`	Integration API	Salesforce
`NWB_Support_Admin`	Support administration	Salesforce

G.3 Profiles

Name	License
`NWB Integration`	Salesforce Integration
`NWB Minimum Access`	Salesforce

G.4 Roles

API name	Label	Parent role
`NWB_Advisor_North`	Advisor — North	`NWB_Branch_Manager_North`
`NWB_Advisor_South`	Advisor — South	`NWB_Branch_Manager_South`
`NWB_Branch_Manager_North`	Branch manager — North	`NWB_Regional_Director_North`
`NWB_Branch_Manager_South`	Branch manager — South	`NWB_Regional_Director_South`
`NWB_CEO`	Chief executive	—
`NWB_Contact_Center_Agent`	Contact center agent	`NWB_Contact_Center_Manager`
`NWB_Contact_Center_Manager`	Contact center manager	`NWB_CEO`
`NWB_Network_Director`	Network director	`NWB_CEO`
`NWB_Regional_Director_North`	Regional director — North	`NWB_Network_Director`
`NWB_Regional_Director_South`	Regional director — South	`NWB_Network_Director`

Appendix H — Integration endpoints

H.1 Named credentials

API name	Label	Type	Host	External credential
`NWB_CoreBanking`	Core banking API	SecuredEndpoint	api.corebanking.northwind-bank.example	`NWB_CoreBanking_OAuth`
`NWB_DocumentStore`	Document store	Legacy	docs.northwind-bank.example	—

API name	Label	Type	Host	External credential
`NWB_SmsGateway`	SMS gateway	SecuredEndpoint	sms.telco-partner.example	`NWB_SmsGateway_Key`

H.2 External credentials

API name	Label	Protocol
`NWB_CoreBanking_OAuth`	Core banking OAuth (client credentials)	Oauth
`NWB_SmsGateway_Key`	SMS gateway API key	Custom

H.3 Connected apps and authentication providers

API name	Label	OAuth scopes
`NWB_Mobile_Banking`	Mobile banking app	Api, RefreshToken

API name	Label	Type	Registration handler
`NWB_Corporate_IdP`	Corporate identity provider	OpenIdConnect	`SsoRegistrationHandler`

H.4 Allowed domains

Kind	API name	Host
Remote Site	`NWB_Maps`	maps.geo-provider.example
CSP Trusted Site	`NWB_Map_Tiles`	tiles.geo-provider.example

Appendix I — Packages and namespaces

Namespace	Product
`FinServ`	Financial Services Cloud

Appendix J — Traceability to the metadata

Chapter	Metadata analysed
4 General architecture	Applications, Lightning pages, installed packages and namespaces
6 Data architecture	Objects, fields, relationships, record types, validation rules, Custom Metadata
7 Application architecture	Apex classes and triggers, flows, OmniStudio components
8 User interface	Applications, Lightning pages, LWC, Aura, Visualforce, tabs, quick actions
9 Integration architecture	Named and External Credentials, remote sites, connected apps, auth providers, CSP and CORS, Apex REST
10 Security and access	Profiles, permission sets and groups, roles, queues, sharing rules, security settings

Chapter	Metadata analysed
12 Environments, delivery and operations	Scheduled jobs, test classes, code coverage (when connected to the org)

Type	Count
`applications`	2
`authproviders`	1
`classes`	18
`connectedApps`	1
`cspTrustedSites`	1
`customMetadata`	5
`dashboards`	1
`email`	1
`externalCredentials`	2
`flexipages`	3
`flows`	4
`groups`	1
`labels`	1
`lwc`	5
`messageChannels`	1
`namedCredentials`	3
`objects`	46
`omniDataTransforms`	1
`omniIntegrationProcedures`	1
`omniScripts`	1
`permissionsetgroups`	3
`permissionsets`	5
`profiles`	2
`queues`	2
`quickActions`	1
`remoteSiteSettings`	1
`reports`	3
`roles`	10
`settings`	1
`sharingRules`	1

Type	Count
`staticresources`	1
`tabs`	4
`triggers`	2

Appendix K — Glossary

Term	Definition
Branch	Unit of the bank network that owns customers and leads by default (Branch__c).
Custom Metadata	Deployable configuration records, read by code and flows without consuming queries.
External Credential	Authentication definition (protocol, principals) used by named credentials.
Flow	Declarative automation of the Salesforce platform (record-triggered, scheduled, screen, autolaunched).
loan application	Request for a consumer loan or a mortgage, from intake to decision (LoanApplication__c).
LWC	Lightning Web Component: standards-based UI component of the Lightning platform.
Named Credential	Secure definition of a callout endpoint and its authentication.
OWD	Organization-Wide Defaults: baseline record access for each object.
PSG	Permission Set Group: bundle of permission sets assigned as one persona.
Queue	Shared owner for records awaiting assignment.
Record Type	Variant of an object with its own business process, picklist values and page layout.